

IAN BURKE

Software Engineer · AI Integration & Automation

ianjamesburke@gmail.com ianjamesburke.github.io github.com/ianjamesburke

Software engineer building AI tools for university staff and video production teams.

EXPERIENCE

Davenport University Current
Automation & AI Integration Specialist

- Built Pounce, a Google Chat agent that helps staff create status reports and donor briefings, submit bugs and ideas, and create Google Docs in shared drives. Built with Next.js, SQLite, Gemini, and locally hosted Gemma and Llama models through LiteLLM.
- Built APIs and MCP servers for university data, including sports search across roughly 282,000 images on campus storage and donor research using Blackbaud NXT. Connected these tools to Pounce, with donor research returned in Google Chat cards and Docs.
- Configured a LiteLLM gateway with separate model keys and spending limits for each app, plus a five-node Gemma pool serving multiple campus agents.
- Built pull-based deployments using systemd and Docker Compose for campus servers that cannot accept inbound connections. Added automated checks to test agent tools after deployment.

Narrative Ads 2022 - Present
Full Stack Developer, NarratorAI

- Built and maintain NarratorAI, an internal platform for AI media production and ads, using TypeScript, Next.js, and Firebase. Brought workflows previously spread across Frame.io, ClickUp, and Google Drive into one application.
- Built image, video, and voiceover generation pipelines using Gemini, fal.ai, ElevenLabs, and OpenRouter.
- Added chat tools that let staff request media generation, review work, and run video-editing jobs.
- Debugged generation jobs, provider failures, and backend capacity issues under load.
- Built tools for querying ad performance data and reconciled monthly costs across GCP, fal.ai, OpenRouter, and Fly.io.

SELECTED PROJECTS

Kill Tony Archive – Built a pipeline to transcribe and extract structured data from 1,422 sets across 110 episodes. Used Python, Gemini, SQLite, FastAPI, and SvelteKit.

PLEXI – Built a Rust desktop workspace for coding agents, with terminal panes, command-line agents, and a WASM app runtime. Available as a macOS prerelease.

SKILLS

Python, TypeScript, JavaScript, Rust · React, Next.js, SvelteKit, FastAPI · Firebase, GCP, SQLite, Docker Compose, systemd · LiteLLM, OpenRouter, MCP, agent tool calling, structured extraction, CI/CD